

Assignment for Submission

Problem 1: A proton is confined in an infinite potential well of length 10 femto-meter. Calculate the energy and wavelength of the photon, when proton undergoes the transition from second excited state to ground state?

Problem 2: An electron is described by a function

$$\Psi(x) = 0 \text{ for } x < 0$$

$$\Psi(x) = Ae^{-x}(1 - e^{-x}) \text{ for } x \geq 0$$

- (i) Normalize the wave function and determine A?
- (ii) Find the position of the particle where particle probability is largest?

Problem 3: Find the energy of an electron of an atom in a stationary state for which the wave function is defined as $\Psi(x) = A(1 + ax)e^{-\alpha x}$, where A, a, and α are constants?

Problem 4: Consider a potential box of length l , having boundaries as $V(x) = 0, 0 < x < l$ and $V(x) = \infty$, elsewhere. Suppose a particle of mass m , is trapped within this box. (i) Estimate the energies of ground state, first excited state, and second excited state, given that particle mass $m = 9.1 \times 10^{-31} \text{ kg}$ and the box length $l = 0.1 \text{ nm}$, (ii) Estimate the energies of ground state, first excited state and second excited state for a particle of mass $m = 1 \times 10^{-3} \text{ kg}$ and the box length $l = 10 \text{ cm}$ (iii) discuss the importance of the quantum effects for both of these systems.

Problem 5: A photon of energy 3 keV collide elastically with an electron initially at rest. If the photon emerges at an angle 60° , calculate (i) kinetic energy of recoiling electron. (ii) the angle at which the electron recoils.

Problem 6: Applying the uncertainty principle estimate the minimum kinetic energy of the electron confined within a region whose size is $l = 0.2 \text{ nm}$?

Problem 7: An electron having kinetic energy $K.E. = 4.0 \text{ eV}$ is confined within a region whose linear dimension is $l = 1.0 \mu\text{m}$. Estimate the uncertainty in its velocity?

Problem 8: In Hall effect measurements a plate of width $w = 10 \text{ mm}$, length $l = 50 \text{ mm}$, made of p-type semiconductor was placed in a magnetic field $B = 5.0 \text{ kGauss}$. A potential difference $V = 10.0 \text{ V}$ was applied across the edge of the plate. In this case the Hall field is $V_H = 50.0 \text{ mV}$, and resistivity $\rho = 2.5 \Omega - \text{cm}$. Find the holes concentration and holes mobility.

Problem 9: Calculate the interval (in eV) between neighbouring levels of free electrons in a metal at $T = 0 \text{ K}$ near the Fermi Level, the concentration of free electrons is $n = 2.0 \times 10^{22} \text{ cm}^{-3}$ and the volume of the metal is $V = 1.0 \text{ cm}^3$.

Problem 10: Find the London penetration depth $\lambda_L(0) = \sqrt{\frac{m_e}{\mu_0 e^2 n_s}}$ at 0 K for lead if the number density of the electron is $n_s = 3.29 \times 10^{28} \text{ m}^{-3}$. Given that $\mu_0 = 4\pi \times 10^{-7} \text{ Tesla m/A}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ Coulomb}$

If $\lambda_L(T) = \lambda_L(0) \left(1 - \left(\frac{T}{T_c}\right)^4\right)$, calculate $\lambda_L(T)$ at 3.61 K from its value at $\lambda_L(0)$, where critical temperature $T_c = 7.22 \text{ K}$.